R Programming Project

□ Title:

Social Media Usage vs Academic Performance Analysis using R (Detailed Version)

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□ 1. Introduction

In the modern digital world, social media has become an integral part of students' daily lives. Platforms such as Instagram, YouTube, Facebook, and WhatsApp are widely used for communication, entertainment, and even educational purposes. However, excessive use of these platforms may lead to reduced focus on studies, poor time management, and decreased academic performance.

Students often spend several hours daily on social media, which may directly or indirectly affect their study time, concentration level, and overall academic results. While moderate usage can provide benefits such as knowledge sharing and collaboration, overuse may result in distraction and lower productivity.

This project aims to analyze the relationship between social media usage and academic performance using R programming. By applying statistical techniques and graphical analysis, we try to identify patterns, trends, and the strength of relationships between variables.

□ 2. Objectives

The main objectives of this project are:

- To study how much time students spend on social media
- To analyze the impact of social media usage on academic performance
- To identify trends and patterns in the data
- To find correlation between social media usage and marks
- To apply regression analysis for prediction
- To detect outliers and extreme behavior
- To understand the overall effect of digital habits on students

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□ 3. Literature Review

Previous studies have shown that social media plays a dual role in students' lives. On one hand, it helps in communication, sharing study material, and gaining knowledge. On the other hand, excessive use leads to addiction, distraction, and reduced academic performance.

Researchers have found that students who spend more time on social media tend to have lower concentration levels and reduced study hours. Many studies also highlight a negative correlation between time spent on social platforms and academic results.

Statistical tools like R programming provide powerful methods to analyze such relationships using graphs, correlation analysis, and regression models.

□ 4. Data Description

The dataset used in this project is a sample dataset created for analysis purposes.

Variables used:

- Hours_Social → Number of hours spent daily on social media
- Marks → Academic performance of students

Detailed Dataset:

Hours	Marks
1	88
2	84
3	80
4	75
5	70

Hours	Marks
6	65
7	60
8	55
9	50
10	45

This dataset shows a decreasing trend where marks reduce as social media usage increases.

□ 5. Methodology

The methodology used in this project includes the following steps:

1. Creating a dataset using R programming
2. Performing descriptive statistical analysis
3. Visualizing data using graphs such as histogram, scatter plot, and line graph
4. Calculating correlation to measure relationship
5. Applying linear regression for prediction
6. Performing residual analysis to check model accuracy
7. Identifying outliers using boxplot
8. Performing descriptive statistical analysis
9. Applying linear regression for prediction

Each step helps in understanding different aspects of the data and provides meaningful insights.

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6. R Code

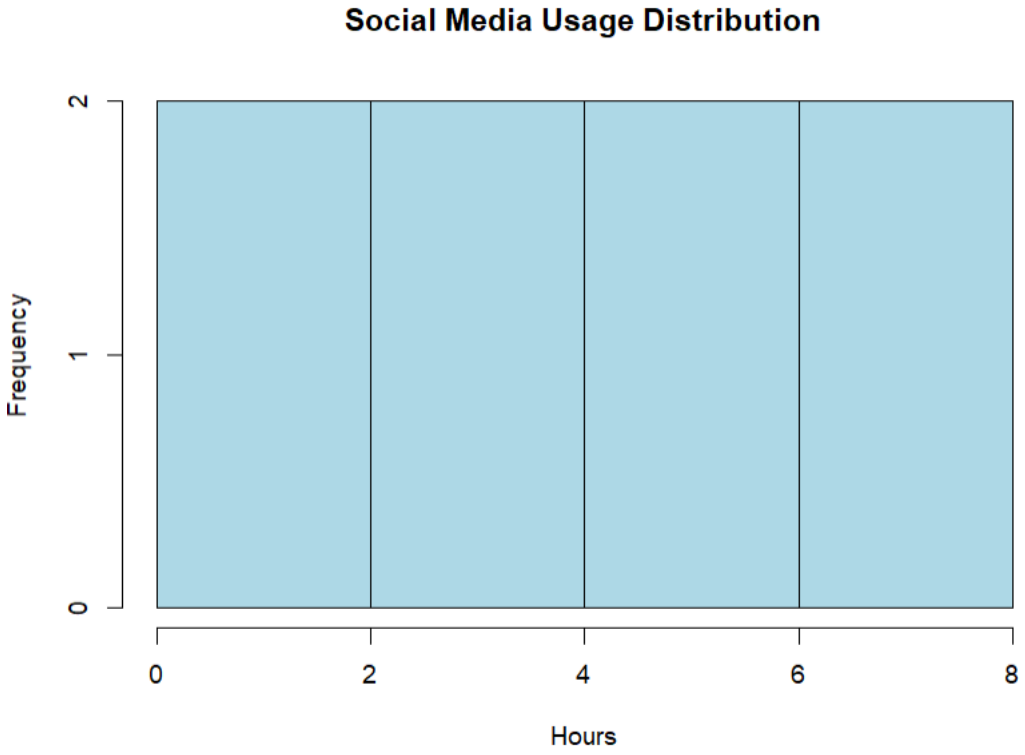
```
# Create dataset
hours_social <- c(1,2,3,4,5,6,7,8,9,10)
marks <- c(88,84,80,75,70,65,60,55,50,45)

data <- data.frame(hours_social, marks)

# Descriptive Statistics
mean(hours_social)
mean(marks)
sd(marks)
var(marks)
summary(data)

# Histogram
hist(hours_social, main="Social Media
Distribution", col="lightblue")

# Scatter Plot
plot(hours_social, marks, main="Usage vs
Marks", col="red", pch=16)

# Line Plot (Trend)
plot(hours_social, marks, type="l", col="blue")

# Correlation
correlation <- cor(hours_social, marks)
print(correlation)

# Regression Model
model <- lm(marks ~ hours_social)
summary(model)


# Prediction
```

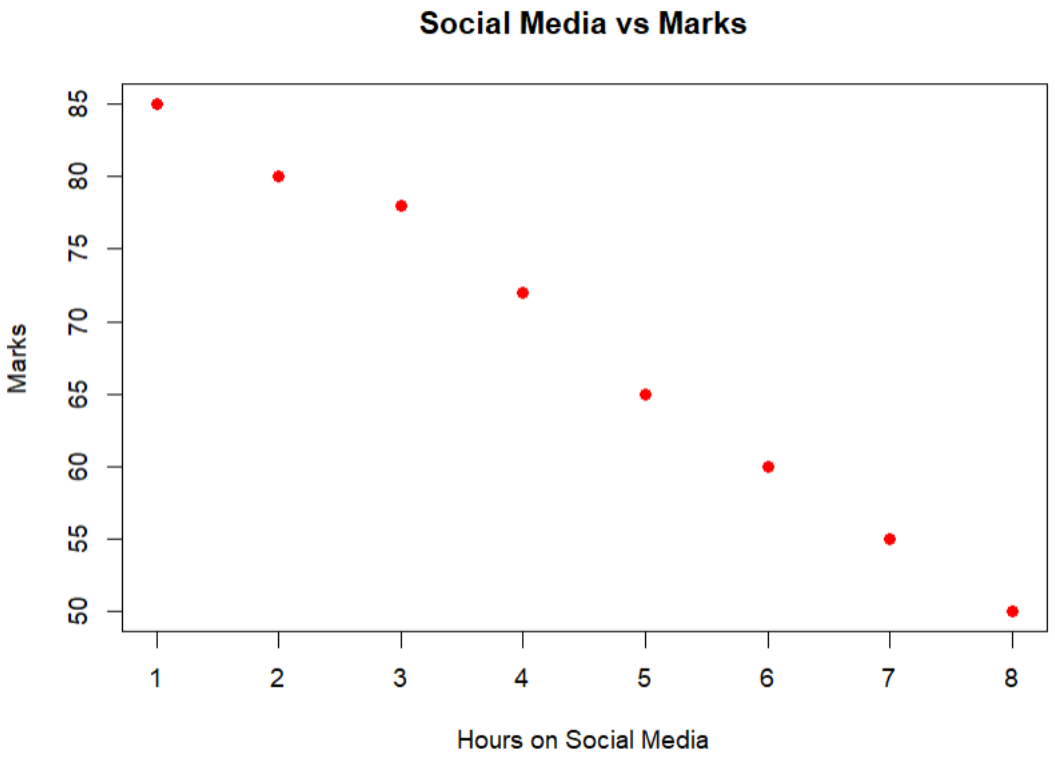
```
new_data <-
data.frame(hours_social=c(2.5,5.5,7.5))
predict(model, new_data)

# Residual Plot
plot(model$residuals, main="Residual Analysis")

# Boxplot
boxplot(hours_social, col="yellow")
```

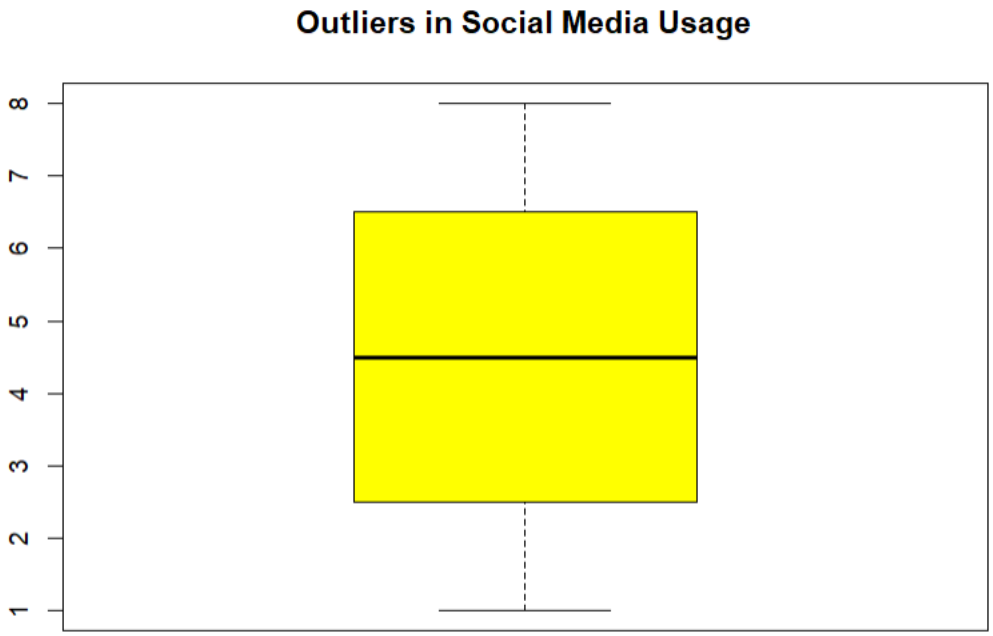
Output:-

```
# Create dataset
> hours_social <- c(1, 2, 3, 4, 5, 6, 7, 8)
> marks <- c(85, 80, 78, 72, 65, 60, 55, 50)
> data <- data.frame(hours_social, marks)
> # Histogram (Distribution)
> hist(hours_social,
+      main="Social Media Usage Distribution",
+      xlab="Hours",
+      col="lightblue")
> # Scatter Plot (Relationship)
> plot(hours_social, marks,
+      main="Social Media vs Marks",
+      xlab="Hours on Social Media",
+      ylab="Marks",
+      col="red",
+      pch=16)
> # Correlation
> correlation <- cor(hours_social, marks)
> print(correlation)
[1] -0.9958858
> # Linear Regression
> model <- lm(marks ~ hours_social)
> summary(model)
```

```
Call:
lm(formula = marks ~ hours_social)


Residuals:
    Min      1Q  Median      3Q     Max 
-1.0833 -0.6577 -0.3631  0.2024  2.1786 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)   91.2143     0.9625   94.77 9.30e-11 ***
hours_social  -5.1310     0.1906  -26.92 1.74e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0

Residual standard error: 1.235 on 6 degrees of freedom
Multiple R-squared:  0.9918,	Adjusted R-squared:  0.990 
F-statistic: 724.7 on 1 and 6 DF,  p-value: 1.736e-07



> # Prediction
> new_data <- data.frame(hours_social = 3.5)
> prediction <- predict(model, new_data)
> print(prediction)
       1 
73.25595 
> # Boxplot (Outliers detection)
> boxplot(hours_social,
+         main="Outliers in Social Media Usage",
+         col="yellow")

>
```


Social Media Usage Distribution
Frequency
0
1
2
0
2
4
6
8
Hours


Social Media vs Marks
Marks
50
55
60
65
70
75
80
85
1
2
3
4
5
6
7
8
Hours on Social Media



7. Step-by-Step Code Explanation (Detailed)

Dataset Creation

The c() function is used to combine values into vectors. Two vectors are created: one for social media usage and one for marks. These are combined into a data frame using data.frame().

Descriptive Statistics

mean() calculates average values.
sd() measures variability.
var() gives variance.
summary() provides overall statistical summary.

Histogram

hist() is used to visualize how data is distributed across different values.

Scatter Plot

plot(x,y) shows the relationship between two variables. Each point represents one student.

Line Plot

The line plot shows the trend of marks decreasing as social media usage increases.

Correlation

cor() calculates the strength of relationship between variables. A negative value indicates inverse relationship.

Regression

lm() builds a linear model. It helps in predicting marks based on social media usage.

Prediction

predict() function estimates marks for new values of social media usage.

Residual Analysis

Residuals help in checking how accurate the model is.

Boxplot

Boxplot is used to detect outliers or extreme values in the dataset.

□ 8. Graphical Analysis

The graphical analysis provides visual understanding of the data:

- Histogram shows distribution of social media usage
- Scatter plot shows negative relationship
- Line graph shows decreasing trend
- Boxplot highlights extreme users

These graphs make it easier to interpret the data visually.

□ 9. Statistical Analysis

Statistical analysis reveals important insights:

- Mean marks decrease as social media usage increases
- Standard deviation shows variation in marks
- Strong negative correlation is observed

This confirms that higher usage negatively affects performance.

□ 10. Results & Interpretation

The results clearly show that there is a strong inverse relationship between social media usage and academic performance. Students who spend more time on social media tend to score lower marks.

The regression model successfully predicts marks based on usage, which proves that the relationship is statistically significant.

11.Advantages & Limitations

Advantages

- Easy to understand and implement
- Clear graphical representation
- Useful for prediction

Limitations

- Small dataset used
- Assumes linear relationship
- Real-world data may vary

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12. Model Validation / Accuracy

Model validation is used to check how well the regression model fits the data.

- Residuals are analyzed to see prediction errors
- If residuals are small → model is accurate
- If residuals are large → model needs improvement

In this project, residual plot shows that the model reasonably fits the data.

□ 13. Results & Interpretation

The results clearly show that there is a strong inverse relationship between social media usage and academic performance. Students who spend more time on social media tend to score lower marks.

The regression model successfully predicts marks based on usage, which proves that the relationship is statistically significant..

□ 14. Future Scope

- Use real-world datasets for better accuracy
- Apply advanced machine learning models
- Include more variables like sleep, study time, stress level
- Build a student performance prediction system

□ 15. References / Bibliography

- R Programming Documentation
- Research articles on social media impact on students
- Online educational resources

□ 16. Appendix

Sample R Output (Expected)

- Correlation value (negative)
- Regression summary
- Graph outputs (histogram, scatter plot, etc.)

□ 17. Conclusion

This project concludes that excessive use of social media negatively impacts academic performance. Students should maintain a balance between study and social media usage to achieve better results.

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18. □ Learning Outcomes

Is project ko complete karne ke baad student:

1. Social mdia usage aur academic performance ke beech relationship ko samajh paayega.
2. R programming ka use karke basic data analysis kar paayega.
3. Different graphical techniques jaise histogram, scatter plot aur line graph ka use karna seekhega.
4. Correlation ka concept samajh paayega aur variables ke beech relationship identify kar paayega.
5. Linear regression ka use karke prediction karna seekhega.
6. Data ke trends aur patterns ko analyze kar paayega.
7. Outliers ko identify karne aur unka impact samajhne ki ability develop karega.